

PROJECT REVIEW PRESENTATION

Predictive Analytics for Diabetes Risk & Health Indicators

Exploratory Data Analysis and Machine Learning Preprocessing Pipeline

- **Project Title:** Predictive Analytics for Diabetes Risk and Health Indicators
- **Team Name:** Team II
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- **Review Date:** 02-08-2026

Project Overview: AI Decision-Support for Early Diabetes Screening

PROBLEM STATEMENT

- Diabetes is a leading cause of cardiovascular disease, kidney failure, & vision loss.
- Clinical lab screening requires expensive specialist visits & lab tests.
- Non-invasive survey data offers an unexploited early screening opportunity.

CORE OBJECTIVE

- Develop an accurate machine learning classification model (Diabetic vs Non-Diabetic).
- Identify key health & lifestyle predictors influencing diabetes risk.
- Provide a reliable decision-support tool for healthcare screening.

PROJECT SCOPE

- CDC BRFSS dataset: 70,692 records with 22 demographic & health attributes.
- Cross-validation against 2014 BRFSS CDC Codebook variables..
- Pipeline scope: Preprocessing, WHO feature engineering, EDA, & ML prep.

EXPECTED OUTCOME

- Validated ML preprocessing pipeline with 100% data integrity.
- Early risk identification tool assisting healthcare professionals.
- Actionable insights on key lifestyle & clinical risk factors.

What We Have Done: Completed Engineering & Pipeline Milestones

01

Feature Engineering: WHO BMI Classification & Binned Health Indicators

- Engineered WHO BMI categories: Underweight (<18.5), Normal (18.5-25), Overweight (25-30), Obesity I (30-35), Obesity II (35-40) and Obesity III (40+).
- Binned continuous health metrics: MentHlth_Group & PhysHlth_Group into 0, 1-5, 6-10, 11-20, 21-30 days.

02

Data Audit & Research: CDC Codebook Verification & Data Inspection

- Audited 70,692 rows × 22 columns; confirmed 0 null/missing entries across all attributes.
- Cross-verified unique value distributions against CDC BRFSS codebook standards.

03

Design & Mapping: Clinical Label Standardization & Categorization

- Mapped 14 binary health indicators (HighBP, HighChol, Smoker, Stroke, etc.) to human labels.
- Standardized 7 ordinal demographic attributes (Age 18-80+, Income \$10k-\$75k+, Education).

04

Pipeline Setup: Environment Configuration, Exploratory Data Analysis and Data Preprocessing

- Configured Python stack: pandas, numpy, seaborn, scikit.
- Mounted Google Drive storage & ingested dataset; built EDA structure for distribution analysis.
- Done preprocessing steps like feature engineering, data splitting, feature selection.

Why We Did Those: Technical Rationale & Business Justifications

Task	Reason	Benefit
WHO BMI Class Discretization	BMI was the sole continuous metric among 21 discrete/ordinal survey features.	Enables non-linear WHO obesity risk group modeling & categorical EDA visual clarity..
Health Days Binned Grouping	Raw MentHlth & PhysHlth ranged continuously from 0 to 30 days.	Reduces response variance noise & uncovers distinct quality-of-life risk bands..
Human-Readable Label Mapping	Raw dataset used arbitrary numeric codes (Age 1-13, Income 1-8).	Eliminates interpretation errors during EDA & improves stakeholder communication.
Unique Value Audit Across Features	Validate data integrity against official CDC BRFSS codebook specs.	Guarantees 100% clean data with zero missing values or invalid codes prior to ML.
ML Pipeline Library Pre-configuration	Prepare for feature scaling, mutual info selection, SelectKBest.	Establishes a standardized, reproducible baseline for model benchmark testing.

Key Inferences & Learnings: Dataset Structure & Analytical Insights

RESULTS OBTAINED & INSIGHTS

- Dataset balance: Exact 50/50 split with 35,346 diabetic/prediabetic & 35,346 non-diabetic records.
- Clinical baseline: High BP (56.3%), High Chol (52.6%), Physical Activity (70.3%), Veggie Intake (78.9%).
- Key predictors: High BP, High Chol, GenHlth (mean 2.84/5), Age (mean cat 8.58 $\approx$ 55-60 yrs), & DiffWalk (25.3%).

PERFORMANCE & MODEL STATUS

- Data Quality: 100% complete dataset verified across 70,692 rows $\times$ 22 columns with 0 missing values.
- Model Metrics: Exact model accuracy, precision, recall, & ROC-AUC are Not Specified in Provided Documents (training pending).
- Pipeline Readiness: Preprocessing workflow validated and fully ready for model training execution.

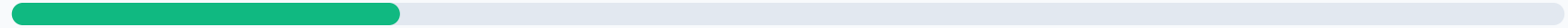
CHALLENGES, SOLUTIONS & LESSONS

- Challenge: Single continuous feature (BMI) amid discrete survey metrics created visual & scaling contrast.
- Solution: Applied standard WHO BMI cutoffs (<18.5 to $\geq$ 40) for categorical alignment.
- Lesson Learned: Upfront schema verification against official CDC codebooks prevents downstream model errors.

Current Progress: Data Pipeline Status & Completion Tracking

Project Progress

25% Done



Data Prep, EDA & Preprocessing (Complete) Model Training & Hosting (Pending)

COMPLETED TASKS (100% DONE)

- ✓ Problem definition & scope formulation.
- ✓ Dataset loading & memory structure inspection.
- ✓ Missing value audit (0 nulls found).
- ✓ Unique value verification across all 22 features.
- ✓ WHO BMI feature discretization (`BMI\_Class`).
- ✓ Human-readable dictionary label mapping.
- ✓ Feature and Target Splitting.
- ✓ Train Test Split.
- ✓ Scaling.
- ✓ Feature Selection.

UP NEXT (PLANNED)

- ML Classifier training (Logistic Reg, Random Forest, XGBoost).
- Fine Tuning & Hyperparameter Optimization.
- Model evaluation, ROC-AUC benchmarking & report docs.
- Website hosting & UI deployment.
- Final Review & Stakeholder Presentation.

THANK YOU